

求积分

$$\begin{aligned}&\int_0^\pi e^{\cos x} \cos \sin x \mathrm{d} x \\&= \frac{1}{2} \int_0^\pi e^{\cos + i \sin x} + e^{\cos - i \sin x} \mathrm{d} x\end{aligned}$$

由奇偶性得到

$$= \frac{1}{2} \int_{-\pi}^\pi e^{e^{ix}} \mathrm{d} x$$

令 $e^{ix} = z$ 得到

$$\begin{aligned}&= -\frac{i}{2} \oint_{|z|=1} \frac{e^z}{z} \mathrm{d} z \\&= \pi Res[\frac{e^z}{z}, 0] = \pi\end{aligned}$$